

# Lab Notebook Entry: No-Refolding, Relational Proliferation, and a Dark-Energy Interpretation in HSF

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## Project Context

This note records a conceptual development within the Hilbert Substrate Framework (HSF). The purpose is not to claim a finished derivation, but to sharpen a mechanism that may connect subsystem proliferation, the no-refolding constraint, and an effective cosmological expansion term interpretable as dark energy.

## Prompting Idea

The core thought is this:

When two subsystems interact, the resulting entanglement is not properly a property of either subsystem independently. If that relational information is irreducible, then it must itself be represented somewhere in the substrate. In HSF language, that means interaction can generate a new subsystem.

This is not being proposed as generic noise or background churn. It is a specific closure rule: interaction creates irreducible relational structure, and irreducible relational structure demands representation.

## Proposed Mechanism

Suppose subsystems  $A$  and  $B$  interact. After interaction, there exists relational information that cannot be attributed to  $A$  alone or  $B$  alone. Denote the corresponding relational subsystem by

$$R_{AB}.$$

The claim is that  $R_{AB}$  is not optional bookkeeping. It is a genuine substrate element required to carry the new relation.

Once  $R_{AB}$  exists, further interactions become possible:

$$A \leftrightarrow R_{AB}, \quad B \leftrightarrow R_{AB}.$$

These interactions can produce further irreducible relational subsystems. Very schematically:

$$(A, B) \rightarrow R_{AB},$$

$$(A, R_{AB}) \rightarrow R_{A,AB}, \quad (B, R_{AB}) \rightarrow R_{B,AB},$$

and so on.

This suggests a proliferative process driven not by arbitrary splitting, but by *interaction-induced relational closure*.

## Role of No-Refolding

The no-refolding constraint is what makes this cumulative rather than trivial.

A rough statement of the principle is:

Once structure exists, the substrate cannot freely recoarse-grain or refactor itself back into fewer, larger subsystems without paying a physical cost.

Without no-refolding, the substrate could respond to every newly generated relation by simply regrouping the total Hilbert space into a coarser factorization. In that case, relational proliferation would never accumulate; it would be continually reabsorbed.

With no-refolding, that easy escape is blocked. Newly generated relational subsystems cannot simply be erased by a cheap redescription. They persist as committed structure.

So the logic becomes:

1. Interaction creates irreducible relational information.
2. Irreducible relational information requires its own subsystem.
3. No-refolding prevents free collapse back to a coarser factorization.
4. Therefore relational subsystems accumulate.

This is the key conceptual bridge.

## Interleaving Rather Than Edge Growth

A major refinement is that this proliferation need not occur only at the “edge” of an already-formed region. The more natural HSF picture is that the new subsystems interleave among existing ones.

That is, if  $A$  and  $B$  already exist in the substrate, and their interaction produces  $R_{AB}$ , then  $R_{AB}$  is not necessarily placed outside the preexisting relational structure. It may instead be inserted *between* or *among* the subsystems already present.

So this mechanism suggests:

The substrate does not expand merely by adding new degrees of freedom at a boundary. It expands by becoming more finely and irreversibly articulated within its interior.

This is important because it gives a natural route to distributed expansion rather than center-outward expansion.

## Dark-Energy Interpretation

The speculative interpretation is then:

Dark energy may correspond, at the HSF level, to cumulative no-refolding-protected interleaving of new relational subsystems generated by interaction.

In this picture, cosmic expansion is not fundamentally “space stretching” in an already-given background. Instead, emergent geometry enlarges because more irreducible relational structure is forced into the substrate over time.

The effective macroscopic effect would be:

- persistent growth of accessible relational depth,
- increased effective separation among previously existing structures,
- a distributed expansion not localized to a boundary,
- a substrate-wide tendency against recoarse-graining.

This is why the mechanism feels dark-energy-like rather than merely growth-like.

## Candidate Principle Statement

A sharper internal HSF statement might be:

**Relational Proliferation Principle.** Whenever interaction generates information that is not attributable to either participant independently, the substrate must represent that information as a distinct relational subsystem. Under no-refolding, such subsystems cannot in general be freely reabsorbed into a coarser factorization, so interaction drives cumulative internal proliferation of relational structure.

And the cosmological corollary would be:

**Dark-Energy Corollary (speculative).** If interaction-generated relational subsystems accumulate throughout the substrate and interleave among existing structures, then the accessible emergent geometry can expand everywhere at once. At large scales this may appear as a dark-energy-like contribution to cosmic expansion.

## What This Would Need to Mean Operationally

For this idea to move from conceptual note to model, it must be tied to observables. The key distinction is:

$$\textit{subsystem count increase} \neq \textit{metric expansion}$$

So the missing bridge is an operational quantity showing that interleaving proliferation increases effective geometric separation.

Candidate observables:

1. **Average graph distance** between persistent structures.
2. **Minimal communication delay** between fixed subsystem identities.
3. **Shortest-path growth** in the emergent interaction graph.
4. **Accessible embedding scale** under the currently allowed factorization.
5. **Recoarse-graining resistance**: a measure of how much physical cost is required to absorb proliferated relations back into larger effective subsystems.

If these increase with relational proliferation, then the claim becomes physically sharper:

the same old structures become farther apart in the emergent geometry because new irreducible relational structure has been forced in between them.

## Immediate Theoretical Tension

This mechanism has an obvious danger: combinatorial blow-up.

If every interaction necessarily creates a persistent new subsystem, and every new subsystem can interact with all existing ones, then subsystem count may explode too quickly to yield anything like a stable universe.

So the framework likely needs one or more regulating conditions:

- not every interaction-generated relation persists,
- finite bandwidth limits which relations can become active subsystems,
- no-signaling restricts which interactions are dynamically reachable,
- many relational subsystems may remain transient,
- some relations may coarse-grain collectively rather than individually.

So the refined picture is not:

every possible relation becomes a permanent subsystem.

It is more likely:

interactions generate candidate relational subsystems, and no-refolding prevents easy erasure of the subset that become physically committed.

## Connection to Matter

This may also help separate cosmological expansion from particle formation.

In the present picture:

- **Expansion-like behavior** comes from accumulation of interleaved relational structure.
- **Matter-like behavior** would come from stable, self-maintaining relational knots within that proliferating background.

So a proton-like object would not be “just more proliferation.” It would be a special case where proliferated relational structure closes into a stable, persistent, localized bound complex.

This suggests a possible hierarchy:

$$\text{interaction} \rightarrow \text{relation} \rightarrow \text{relational subsystem} \rightarrow \begin{cases} \text{transient proliferative background} \\ \text{persistent bound structure} \end{cases}$$

That may become important later when trying to define mass operationally.

## Current Status

At present, this is a conceptual mechanism, not a completed derivation. It does not yet show:

- a formal growth law for relational subsystem proliferation,
- an explicit metric observable derived from that proliferation,
- a late-time effective equation of state,
- or a quantitative match to cosmological dark energy.

What it *does* provide is a coherent HSF-native hypothesis:

Cosmic expansion may arise not from external stretching or edge accretion, but from interior accumulation of irreducible relational structure. No-refolding prevents this structure from being cheaply reabsorbed, so interaction drives ongoing interleaved growth of emergent geometry.

## Working Conclusion

The idea worth preserving is:

**In HSF, dark-energy-like expansion may be the macroscopic signature of no-refolding-protected relational proliferation.**

More explicitly:

Entangling interactions generate irreducible relational information. HSF requires that such information be represented by distinct relational subsystems. No-refolding prevents those subsystems from being freely reabsorbed into coarser factorizations. Their cumulative interleaving within existing structure can therefore drive an effective expansion of emergent geometry.

## Next Steps

Concrete next tasks:

1. Define a toy growth rule for interaction-generated relational subsystems.
2. Distinguish transient relational offspring from persistent committed ones.
3. Define an effective distance observable on the resulting subsystem graph.
4. Test whether interleaving proliferation increases that distance over time.
5. Check whether no-refolding is necessary for cumulative growth.
6. Explore whether the coarse-grained expansion law resembles a cosmological source term.

## Notebook Summary

This entry records a shift in interpretation:

- The proposed source of expansion is *not* generic churn.
- The proposed source is *interaction-generated relational subsystem proliferation*.
- The persistence mechanism is *no-refolding*.
- The geometric effect is *interleaved interior growth*, not mere edge accretion.
- The cosmological interpretation is *a possible HSF origin for dark-energy-like expansion*.

That is the current conceptual state of the idea.